

Hurricane-Proofed Skyscrapers No Match for Derecho Winds

A new study in *Frontiers in Built Environment* has found that not all winds are created equal when it comes to damage to structures.

On May 16, 2024, a downburst in Houston damaged multiple skyscrapers in the downtown area of the city, causing extensive damage. These buildings had been constructed to withstand the impact of hurricane-force winds, but the derecho, which had lower wind speeds, caused more damage than any recent hurricane.

Downbursts are strong downward winds that blow outward in all directions once they hit the ground—and the reason why winds are often much more intense around the ground floor of tall buildings.

For the study, the authors analyzed the impact of the May derecho on five iconic buildings in Houston: the Chevron Building Auditorium, the CenterPoint Energy Plaza, the El Paso Energy Building, the RRI Energy Plaza, and the Wedge International Tower. Built between 1962 and 2003, these high-rises are 158 to 226 meters tall. All conformed to the construction standard which dictates that tall buildings be designed to withstand winds up to 67 meters per second, corresponding to a category 4 hurricane.

Wind speeds measured in downtown Houston during the derecho didn't come close to this construction standard, as they peaked at 40 meters per second. Nonetheless, as illustrated by the study, facade panels on these tall buildings were dislodged while cladding panels was damaged, especially on corners and lower floors. Numerous windows cracked or shattered, raining dangerous debris down into the streets.

In contrast, these tall buildings sustained minimal damage during hurricane Beryl, which hit Houston on July 8th, 2024. The maximum wind speed measured in downtown Houston during Beryl was 36 meters per second, comparable to the derecho.

The researchers proceeded to simulate downbursts and hurricanes at the Wall of Wind experimental facility, whose 12 jet fans can generate wind speeds up to 70 meters per



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May 19, 2024 - Houston, Texas: A strong wind blew through downtown and ripped out windows on the Chevron Building in downtown Houston.

second. These were blasted against a revolving miniature representing a tall building on a 1:350 scale. An identical miniature stood at increasing distances ranging from 0.14 to 0.70 meters from the first, to mimic interference from neighboring buildings.

The authors compared two conditions that differed in the variation of the mean wind speed over time: a constant average speed typical of hurricanes and a speed that at first rapidly ramped up, reached a plateau, and then ramped down, characteristic of downbursts. The results showed that there was far more suction on the sides of buildings during downburst events than during hurricanes.

"When strong winds move through a city, they can bounce due to interference between tall buildings. This increases pressure on walls and windows, making damage more severe than if the buildings were isolated," said Omar Metwally, a doctoral student who was the study's first author.

He added, "On top of this, downbursts create intense, localized forces which can exceed typical design values for hurricanes, especially on the lower floors of a tall buildings."

Exposure to High Temperatures Hastens Biological Aging

People in neighborhoods that experience more days of high heat show greater biological aging on average than residents of cooler regions, according to a study by researchers at the University of Southern California.

Biological age is a measure of how well the body functions at the molecular, cellular, and system levels, as opposed to chronological age based on one's birthdate; having a biological age greater than one's chronological age is associated with higher risk for disease and mortality.

Lead author Jennifer Ailshire and colleagues examined how biological age changed in more than 3,600 Health and Retirement Study (HRS) participants aged 56 and older over a six-year study. They then compared participants' changes in biological age to their location's heat index history and number of heat days reported by the National Weather Service from 2010 to 2016.

The NWS Heat Index Chart categorizes heat index values into three levels based on the potential risk of adverse health effects. The "Caution" level includes heat index values ranging from 80°F to 90°F, the "Extreme Caution" level includes values between 90°F and 103°F, and the "Danger" level includes values between 103°F and 124°F. Days in all three levels were included as heat days in the study.

The analysis revealed a significant correlation between neighborhoods with more days of extreme heat and individuals experiencing greater increases in biological age. This correlation persisted even after controlling for socioeconomic and other demographic differences, as well as lifestyle factors such as physical activity, alcohol consumption and smoking.

"Participants living in areas where heat days, as defined as Extreme Caution or higher levels ($\geq 90^\circ\text{F}$), occur half the year, such as Phoenix, Arizona, experienced up to 14 months of additional biological aging compared to those living in areas with fewer than 10 heat days per year," the researchers said. "Even after controlling for several factors, we found this association. Just because you



Desert areas could harvest fog to boost water supplies.

live in an area with more heat days, you're aging faster biologically."

All three epigenetic clocks employed in the study showed the association after short (7 days) and medium (30-60 days) periods of time, indicating that heat-related epigenetic changes could happen relatively quickly, and some of them may accumulate over time.

'Fog Harvesting' Could Boost Water Supplies

Water harvested from fog could prove to be an important resource for water-scarce cities, according to researchers in Chile's Atacama Desert. The desert is one of the driest places on earth, with annual rainfall less than 1 mm per year.

The researchers conducted a year-long field study in Alto Hospicio, a fast-growing municipality located in the Atacama Desert. Because of the city's rapid growth, approximately 10,000 people live in informal settlements. Only 1.6% of these settlements are connected to water distribution networks, and most inhabitants receive water via trucks.

Fog collectors generally consist of a mesh suspended between two posts. The mesh serves as an interception surface to catch moisture. Droplets collate on the mesh and fall into a gutter leading to water storage tanks. It's

a passive system that requires no external energy.

Dr. Virginia Carter Gamberini, an assistant professor at Universidad Mayor and first co-author of the *Frontiers in Environmental Science* study, and colleagues found that in a 100-square-kilometer area surrounding Alto Hospicio, between 0.2 and 5 liters of fog water could be harvested per square meter each day. This potential, however, is confined to the higher lying altitudes outside of the city's limits. During the study's peak season, in August and September of 2024, collection potential reached up to 10 liters per square meter and day.

Based on an annual average water collection rate of 2.5 liters per square meter per day, the researchers said that 17,000 square meters of mesh could produce enough water to meet the weekly water demand (300,000 liters) for informal urban settlements. 110 square meters could meet the annual demand for the irrigation of the city's green spaces (100,000 liters). Similarly, fog water could be used for soil-free agriculture, with yields of 15 to 20kg of leafy green vegetables in a month.

"This research represents a notable shift in the perception of fog water use—from a rural, rather small-scale solution to a practical water resource for cities," said Gamberini. "Our findings demonstrate that fog can serve as a complementary urban water supply in drylands where climate change exacerbates water shortages." W

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